COMP4141 Slides Week 5

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Class Test Last Week

- Some students has $\mathbb{N}$ include 0 and others did not. You should use that $\mathbb{N}$ contains 0.
- This week will just be on proving a languages is not regular.

Notes from marking

- Try your construction on some examples to understand where it might fail if you are not sure (particularly related to the DN task from Week 3).
- Most who did the HD task found it easier than the DN one.

Starts on Marking Moving forwards

- This will have:
  - **0 star:** not acceptable, completely wrong.
  - **1 star:** on the right track, but not sufficient and details missing.
  - **2 stars:** almost there, but somewhat incomplete.
  - **3 starts:** all correct.

Turing Machines Formally

Scope

Church–Turing Thesis

Turing machines define computability.

Definition

Turing machine

A TM is a tuple

$$M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$$

with components:

- Q : set of states
- Σ : input alphabet, not containing the blank symbol $\sqcup$
- $\Gamma \supseteq \Sigma$: tape alphabet, containing $\sqcup$
- $\delta : Q \times \Gamma \to Q \times \Gamma \times \{l, r\}$
- q_0 : start state
- q_{accept} : accept state
- q_{reject} : reject state ($\neq q_{\text{accept}}$)

Configurations

Configuration

A configuration for a TM is a triple

$$(u, q, v) \in \Gamma^* \times Q \times \Gamma^*$$

- q is the current state.
- The tape content is uv, with the head at the first symbol of v (blanks suppressed).

For $a, b, c \in \Gamma$, $u, v \in \Gamma^*$, and $q, q' \in Q$:

- Configuration $uaqbv$ yields $uq'acv$ if $\delta(q, b) = (q', c, l)$.
- Configuration $uaqbv$ yields $uacq'v$ if $\delta(q, b) = (q', c, r)$.

Acceptance in Turing Machines

Turing machine acceptance

- $q_0 w$ is called the start configuration.
- $uq_{\text{accept}}v$ are accepting configurations.
- $uq_{\text{reject}}v$ are rejecting configurations.

A TM $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$ accepts input $w \in \Sigma^*$ if a sequence of configurations $C_1, \ldots, C_n$. exists such that:

1. $C_1 = q_0 w$
2. each C_i yields C_{i+1}
3. C_n is an accepting configuration

The language $L(M)$ of M is the set of strings accepted by M.

Encodings

Encodings

Encodings

The input to a Turing machine is always a string, but it can be useful to work with objects other than words over Σ.
If O is a finitely presented object, an encoding of O (over Σ), written $\langle O \rangle$, is a representation of O as a string.

Universal Turing Machine

Simulating a Turing machine

A universal TM is a TM U that can simulate the actions of any Turing machine.
For any TM M and input x, U accepts (respectively rejects or loops on) $\langle M, x \rangle$ if and only if M accepts (respectively rejects or loops on) x.
This is analogous to an interpreter for C written in C.

Decidability and Recognisability

Recognisability

Turing-recognisable

A language $L \subseteq \Sigma^*$ is Turing-recognisable (or recursively enumerable, semi-decidable) if some TM M recognises it, i.e. $L = L(M)$.

Decidable

A TM can fail to accept some input in two ways:

- rejecting: eventually entering q_{reject}
- diverging: entering an infinite loop

Deciders

We call TMs that never diverge deciders.

Decidable

A language $L \subseteq \Sigma^*$ is (Turing-)decidable (or recursive) if some TM decides it.

Enumerators

An enumerator is a TM-like construct that starts with a blank input tape and, if it does not halt, prints an infinite list of strings onto the tape.
It does so by occasionally entering a print state, which outputs the terminal string to the right of the head (including the symbol under the head).

Recognisability

A language is Turing-recognisable iff some enumerator enumerates it.

Recursively Enumerable vs Countable

Countability

A set is said to be countable if its elements can be listed out (formally, a countable set is either finite or countably infinite; the latter meaning there is a one-to-one correspondence between it and the natural numbers $\mathbb{N}$).

Recursive enumerability for a language is stronger than countability, as it essentially asks us to compute such a list as well.

Summary

L is RE (or semi-decidable) if there is a TM such that:

- For every $w \in L$, the TM accepts w in finite time.
- For every $w \notin L$, the TM either rejects or diverges.

L is recursive (or decidable) if there is a TM such that:

- For every $w \in L$, the TM accepts w in finite time.
- For every $w \notin L$, the TM rejects w in finite time.

W5 P1

Problem 1

Give a detailed description of a Turing machine that decides the language L over alphabet $\{0, 1\}$ consisting of all strings with the same number of 0's as 1's. You may find the simulator here useful to debug your solution.

W5 P2

Problem 2

Let

$$L = \{\langle M \rangle \mid M \text{ is a DFA that doesn't accept any string containing an odd number of 1's}\}$$

Show that L is decidable.

W5 P3

Problem 3

A nondeterministic Turing machine is a *decider* for a language L if for all inputs w, the every branch of the computation tree of the machine terminates, and $w \in L$ iff at least one branch accepts. Show that L is decided by a non-deterministic Turing machine iff L is decided by a deterministic Turing machine.

W5 P4

Problem 4

An *infinite word* over Σ is an infinite sequence $\rho = a_0 a_1 a_2 \ldots$ such that $a_i \in \Sigma$ for all i. Show that the set of infinite words over $\{0, 1\}$ is uncountable.

W5 P5

Problem 5

Show that the language $\{\langle M \rangle \mid M \text{ is a Turing machine with } L(M) \neq \emptyset\}$ is Turing-recognisable.

Feedback

Please let me know how I can improve for next week here!